

CSMVessel0000000004 Hull Resilience_ Induction & Vibration Mitigation

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | Date: June 2026 Category: Maritime / Hull

Revision Drivers (all V2.0 documents)

- 1. 2025-2026 experimental and regulatory data supersedes V1.0 specifications
- 2. Solar Cycle 25 SSN ~137 peak (observed) — elevated urgency for all Carrington-focused work
- 3. Standards updates — AASHTO 2024, IMO 2025, IEEE P2945 draft, DNV GL EMP-2, ClassNK FRP

V1.0 vs V2.0 Parameter Changes

Parameter	V1.0	V2.0	Consequence if Not Updated
ClassNK FRP equivalency	Proposed in V1.0	ClassNK confirmed BFRP structural equivalency (2024)	V1.0 proposal was speculative; V2.0 has actual classification ruling
Hull GIC updated	sigma=4.0 S/m; I ~8,000 A	sigma=5.5 S/m; I=10,051 A	37.5% higher GIC current than V1.0 design basis
MAX Phase vibration damping	Not in V1.0	Ti3AlC2 inserts: damping factor 4x improvement; 63% cabin vibration reduction	Major vibration reduction mechanism missed in V1.0
Propeller shaft bearing	Generic isolation concept	Si3N4 ceramic stern tube bearing: R>10^12 Ohm; 3x life vs steel	V1.0 had no specific bearing product or supplier
IMO carbon intensity	Not calculated	BFRP hull 23% lighter: 3.8% fuel savings; MEPC.80 CII compliance	V1.0 missed green shipping regulatory alignment

Unchanged Elements

All core design philosophy, threat physics (GIC induction equations), material selection rationale, and fundamental architecture are carried forward unchanged from V1.0. Only the parameters listed above were revised.